

2014 exam

Question 1

All of the following statements are false about history of Anatomy except

- A. Anatomy is architecture of the body
- B. The earliest Anatomy was written on papyrus
- C. Anatomy was first studied in Egypt
- D. Cutting apart is the old definition of anatomy
- E. All

Correct Answer: E. All

A. Anatomy is architecture of the body: Historically, pioneered by figures like Andreas Vesalius, anatomy has been metaphorically and academically defined as the "fabric" or "architecture" of the living body.

B. The earliest Anatomy was written on papyrus: The *Edwin Smith Papyrus* (dating back to circa 1600 BC in ancient Egypt) is universally recognized as the oldest surviving document containing scientific anatomical descriptions and clinical observations.

C. Anatomy was first studied in Egypt: The earliest systematic anatomical examinations took place in ancient Egypt, heavily influenced by their embalming and mummification practices.

D. Cutting apart is the old definition of anatomy: The word "anatomy" comes from the Greek roots *ana-* (meaning "apart" or "up") and *temnein* (meaning "to cut"). Dissection or "cutting apart" is the literal, historical definition of the word.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 2

Who is the first scholar in history of Anatomy and use the term "Anatomy" for the first time

- A. Galen
- B. William
- C. Hippocrates
- D. Aristotle

Correct Answer: D. Aristotle

Explanation:

✓ **Aristotle** (384–322 BC) was the first scholar in history to use the Greek term *anatome* (meaning "cutting up" or "dissecting") to describe the structural study of organisms. While Hippocrates is celebrated as the "Father of Medicine" and Galen is famous for his subsequent Roman-era anatomical texts, Aristotle coined and introduced the term itself.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 3

Which one of the following division of anatomy study about structures too small to be seen with a light microscope

- A. Applied Anatomy
- B. Microscopic Anatomy
- C. Embryology
- D. Pathology
- E. Ultrastructural Anatomy

Correct Answer: E. Ultrastructural Anatomy

Explanation:

✓ **Ultrastructural Anatomy** (also known as submicroscopic anatomy) is the branch that studies cellular components and organelles that are far **too small to be resolved using a standard light microscope**, requiring the use of an **electron microscope** instead.

Microscopic Anatomy (Option B) typically refers to histology, which studies cells and tissues that *can* be viewed with a conventional light microscope.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 4

Which one of the following is false concerning the anatomy position

- A. Standing in upright position
- B. Eyes are looking ventral to the vertical plane
- C. Arms are at the lateral
- D. Palms directed anteriorly and lower limbs feet together
- E. None

Correct Answer: E. None

Explanation: In the standardized **anatomical position**, all of the listed criteria are correct:

- ✓ The subject is **standing in an upright position** with feet flat on the floor.
- ✓ The **eyes are looking anteriorly/ventrally** (level with the head, looking straight ahead).
- ✓ The **arms are at the sides (lateral)**.
- ✓ The **palms are directed anteriorly** (forward) and the lower limbs are close together with feet directed forward.

◇ Because all of these statements are anatomically true, **none** of them is false.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 5

Dr. Chala is Anatomist who teaches in SPHMMC in Addis Ababa. He takes Residents of surgery to the dissection room to teach them human anatomy using cadaver. Which one of the following approaches to teach human anatomy he will use in the dissection room?

- A. Systemic
- B. Clinical
- C. Regional
- D. Applied

Correct Answer: B. Clinical

Dr. Chala is teaching surgical residents and using cadavers to teach anatomy in relation to surgery and patient care.

- ◇ Systemic anatomy → studies the body system-by-system (e.g., cardiovascular system).
- ◇ Regional anatomy → studies all structures within a particular body region (e.g., thorax).
- ◇ Clinical anatomy → applies anatomical knowledge to clinical practice, diagnosis, surgery, and patient care.
- ◇ Applied anatomy → uses anatomy in practical/clinical contexts, but in this question the surgical-resident context points to clinical anatomy.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 6

Which one of the following movement is not perpendicular to the floor

- A. Medial rotation
- B. Lateral rotation
- C. Adduction
- D. Pronation
- E. Supination

Correct Answer: C. Adduction

Explanation: This question asks about the relationship of the **axis of movement** to the floor when a person is in the anatomical position:

✓ Rotational movements (such as **medial rotation**, **lateral rotation**, **pronation**, and **supination**) occur around a vertical/longitudinal axis. Since this longitudinal axis runs straight up and down, it is **perpendicular to the floor**.

✓ **Adduction** (and abduction) occurs around an anteroposterior (sagittal) axis. Since this axis runs horizontally from front to back, it is **parallel to the floor** (and therefore **not perpendicular to the floor**).

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 7

Which one of the following is not special type of movement

- A. Abduction
- B. Protrusion
- C. Reposition
- D. Eversion
- E. Elevation

Correct Answer: A. Abduction

Explanation:

✓ **Abduction** is a standard, fundamental angular movement that occurs in the coronal plane at many major synovial joints throughout the body (such as the shoulder and hip).

✓ In contrast, **protrusion** (jaw/scapula), **reposition** (thumb), **eversion** (foot), and **elevation** (scapula/jaw) are classified as **special types of movements** because they are highly unique to specific joints.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 8

Which one of the following is belongs to splanchnic skeleton

- A. Heart skeleton
- B. Teeth
- C. Appendicular
- D. Ligaments

Correct Answer: A. Heart skeleton

Explanation: In comparative osteology and anatomy, the skeleton is broadly divided into axial, appendicular, and splanchnic (visceral) divisions. The **splanchnic skeleton** consists of bones or calcified supportive structures that develop inside the substance of soft visceral organs. A classic example of a splanchnic skeletal structure is the **heart skeleton** (such as the *os cordis* found in the hearts of certain larger mammals).

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 9

All of the followings are pneumatic bone except

- A. Frontal
- B. Sphenoid
- C. Zygomatic
- D. Ethmoid

Correct Answer: C. Zygomatic

Explanation:

✓ **Pneumatic bones** are those that contain air-filled spaces (sinuses or air cells), which serve to reduce weight and add resonance to the voice.

✓ The **frontal**, **sphenoid**, and **ethmoid** bones all contain paranasal sinuses (frontal sinus, sphenoid sinus, and ethmoid air cells respectively).

✓ The **zygomatic** (cheek) bone is a flat/irregular facial bone that does not contain any paranasal air cavities.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 10

Among the terms of bone marking which one of the following term stands for "smooth, flat surface on a bone usually covered with cartilage"

- A. Facet
- B. Capitulum
- C. Process
- D. Malleolus

Correct Answer: A. Facet

Explanation:

◇ In anatomical bone terminology, a **facet** is a smooth, flat articular surface where one bone meets another, typically covered with hyaline cartilage to allow gliding movements (such as the articular facets of vertebrae).

◇ A *capitulum* is a small, globular, or head-like articular projection.

◇ A *process* is a general term for any bony prominence or projection.

◇ A *malleolus* is a large, rounded prominence.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 11

Hussein is 17 years old male patient in Emergency department of Agaro general hospital. He come with chief complain of failing down from motor bicycle and he feels pain on his left leg near pelvic. The X-ray of his left leg shows the head of femur is driven into pelvic bone. Which one of the following terms of bone fracture is used to diagnosis the above case scenario?

- A. Colles fracture
- B. Impacted fracture

- C. Fissured fracture
- D. Greenstick fracture

Correct Answer

: B. Impacted fracture

e

Explanation:

✓ An **impacted fracture** is defined as a fracture where one fragment of bone is forcefully driven into another. In this clinical scenario, the high-force motorcycle accident caused the head of the femur to be driven directly into the pelvic bone (acetabulum), representing a classic impacted fracture.

✓ *Colles' fracture* is a specific distal radius fracture; *greenstick fracture* is an incomplete bend/break typical of young children; and *fissured fracture* is an incomplete longitudinal split.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 12

Which one of the following joint is slightly movable or Amphiarthroses

- A. Syndesmoses
- B. Symphyses
- C. Hip joint
- D. Saddle joint

Correct Answer: B. Symphyses

Explanation:

✓ **Amphiarthroses** are slightly movable joints. **Secondary cartilaginous joints (symphyses)**, such as the pubic symphysis and intervertebral disks, "**are joined by fibrocartilage and are slightly movable joints**".

✓ *Syndesmoses* are classified as fibrous joints (synarthroses), while the *hip joint* and *saddle joint* are synovial joints (diarthroses), which permit free movement.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 13

Which one of the following characteristics of muscles tissue stands for "Ability to respond to a stimulus"

- A. Irritability
- B. Extensibility
- C. Contractility
- D. Elasticity

Correct Answer: A. Irritability

Explanation:

◇ **Irritability** (also referred to as **excitability**) is the physiological characteristic of muscle (and nerve) tissue that defines its **ability to receive and respond to a stimulus** (such as a chemical neurotransmitter or electrical impulse).

◇ *Contractility* (Option C) is the ability to shorten or contract forcibly.

◇ *Extensibility* (Option B) is the ability to stretch or extend without damage.

◇ *Elasticity* (Option D) is the ability to recoil and return to its original resting length after being stretched.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 14

Gastrocnemius is

- A. Fusiform
- B. Convergent
- C. Bipennate
- D. Quadrature

Correct Answer: C. Bipennate

Explanation:

◇ In skeletal muscle architecture, **pennate muscles** have fascicles that attach obliquely to a central tendon. The **gastrocnemius** (the major calf muscle) is a classic example of a **bipennate muscle**, as its muscle fibers run obliquely from both sides of a central tendon structure.

◇ Examples of other muscle shapes include: *Fusiform* (biceps brachii), *Convergent* (pectoralis major), and *Quadrature* (pronator quadratus).

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 17

Fibers conveyed by cranial and spinal or peripheral nerves may be general somatic afferent (GSA) which can be exteroceptive and **proprioceptive** sensation. What is the meaning of bold term?

- A. Temperature
- B. Pain
- C. Touch
- D. Position

Correct Answer: D. Position

Explanation: General somatic afferent (GSA) fibers transmit exteroceptive sensations (such as pain, temperature, and touch) and proprioceptive sensations. **Proprioception** is the medical term for **position sense** (the awareness of the position and movement of the body and limbs in space).

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 18

Autonomic nervous system consists of motor fibers that stimulate smooth (involuntary) muscle & glandular (secretory) cells the efferent nerve fibers & ganglia. Which of the following of the ANS (thoracolumbar) origin?

- A. Sympathetic
- B. Parasympathetic
- C. Craniosacral
- D. All

Correct Answer: A. Sympathetic

Explanation: The autonomic nervous system is divided based on its anatomical outflow. The **sympathetic division** arises from the T1 to L2 spinal cord levels and is referred to as the **thoracolumbar outflow**. Conversely, the parasympathetic division arises from the brainstem and sacral segments (S2–S4), which is the craniosacral outflow.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 20

"Prickly layer" stands for which of the following epidermal stratum or layer

- A. Basale
- B. Spinosum
- C. Granulosum
- D. Lucidum
- E. Corneum

Correct Answer: B. Spinosum

Explanation: In histology and integumentary anatomy, the **Stratum Spinosum** is also known as the "**prickly layer**" (or spinous layer). This name comes from the spine-like or "prickly" appearance of its keratinocytes under microscopy, which is caused by the shrinkage of cells around their abundant desmosomal cell-to-cell connections.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 21

Which of the following is not true?

- A. Regional Anatomy study all structures located in a particular region of the body.
- B. Vertebral column forms part of axial skeleton and it is 33 in young age
- C. The coronal suture separates the frontal bone from the parietal bone
- D. sphenoid bone has cribriform plate

Correct Answer: D. sphenoid bone has cribriform plate

Explanation: This statement is **false** (and therefore the correct answer to the "except" question) because the cribriform plate belongs to the **ethmoid bone**, not the sphenoid bone.

The other options are true:

- ✓ Regional anatomy deals with structural relationships of a particular region of the body.
- ✓ The young vertebral column contains 33 separate vertebrae before sacral and coccygeal fusion occurs.
- ✓ The coronal suture separates the frontal bone from the parietal bones.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 22

All are parts of vertebrae except

- A. Body
- B. Arch
- C. spinous
- D. longitudinal process

Correct Answer: D. longitudinal process

Explanation: A typical vertebra consists of a body, an arch (pedicles and laminae), and several specialized processes. These include the **spinous process**, transverse processes, articular processes (facets), mamillary processes, and accessory processes. There is **no such structure** as a "longitudinal process" on a vertebra.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 23

Which vertebrae has costal facet for origin of ribs

- A. Cervical
- B. Sacrum
- C. Lumbar
- D. Thoracic

Correct Answer: D. Thoracic

Explanation: The thoracic vertebrae are unique in having specialized facets for articulation with the ribs. The "**costal facets or processes of the thoracic vertebrae... articulate with the heads of the corresponding and subjacent (just below) ribs**". Neither cervical, lumbar, nor sacral vertebrae possess these costal facets.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 25

Intrinsic muscles have origin and insertion at the same region

A. False

B. True

Correct Answer: B. True

Explanation: By anatomical definition, **intrinsic muscles** have both their origin and insertion confined within the same specific region (for example, the intrinsic muscles of the hand, like the interossei, or the intrinsic muscles of the tongue).

✓ Conversely, **extrinsic muscles** originate outside of the destination region and insert into it (such as the extrinsic flexor muscles of the hand, which originate in the forearm).

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 26

All are the site of intramuscular drug administration except

A. Deltoid

B. Vastus lateralis

C. Biceps brachii

D. Below umbilicus

Correct Answer: D. Below umbilicus (and C. Biceps brachii)

Explanation:

◇ Standard sites for **intramuscular (IM) drug administration** include the **deltoid muscle** (upper arm) and the **vastus lateralis muscle** (anterolateral thigh, especially preferred in infants).

◇ The abdominal region **below the umbilicus** is a classic site for **subcutaneous injections** (such as insulin or heparin) into the fatty tissue, not intramuscular injections.

◇ Additionally, the **biceps brachii** is not used clinically for IM injections because the

deltoid is the standard muscle of the upper arm, and injecting into the biceps carries a higher risk of injuring major nerves and vessels in the arm.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 27

Heart is located in

- A. Middle mediastinum
- B. Anterior mediastinum**

Correct Answer: A. Middle mediastinum

Explanation: The mediastinum is divided into superior and inferior divisions (with the inferior subdivided into anterior, middle, and posterior). The **heart and its protective pericardial sac** are situated specifically in the **middle mediastinum**.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 28

Which is not true

- A. Red blood cells have 120 days life span
- B. Red blood cells have biconcave disc shape**
- C. Monocyte is categorized under granular leukocyte
- D. Platelets help for blood clotting

Correct Answer: C. Monocyte is categorized under granular leukocyte

Explanation: This statement is **false** (making it the correct answer to the "except" question).

◇ **Monocytes** and lymphocytes are classified as **agranular (nongranular) leukocytes** because they lack conspicuous chemical granules in their cytoplasm.

◇ Granular leukocytes (granulocytes) include neutrophils, eosinophils, and basophils. All other options are true: red blood cells are biconcave discs with a typical 120-day lifespan, and platelets play a key role in clotting.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 29

Which chamber of the heart is most muscular

- A. left atrium
- B. left ventricle**
- C. right ventricle
- D. none

Correct Answer: B. left ventricle

Explanation: The **left ventricle** has a wall that is **two to three times thicker** than that of the right ventricle. It is the most muscular chamber because it performs the hardest work, pumping oxygenated blood throughout the entire high-resistance systemic circulation.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 30

Bicuspid valve is located between
A. left ventricle and aorta
B. right atrium and right ventricle
C. left atrium and left ventricle

Correct Answer: C. left atrium and left ventricle

Explanation: The **bicuspid valve** (also known as the **mitral valve** because of its shape resembling a bishop's miter) is the left atrioventricular valve, located specifically between the **left atrium and the left ventricle**.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 31

Circumduction is a circular movement which involves sequential flexion, abduction, extension and adduction
A. True
B. False

Correct Answer: A. True

Explanation: **Circumduction** is defined as a circular, cone-like movement of a limb or body part. It is a compound movement that sequentially combines **flexion, abduction, extension, and adduction** (or the reverse order) at a biaxial or multiaxial joint.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 32

Which chamber forms base
A. left ventricle
B. left atrium
C. right ventricle
D. none

Correct Answer: B. left atrium

Explanation: The posterior aspect of the heart, known as its **base**, is formed primarily and predominantly by the **left atrium** (and only a small part of the posterior right atrium).

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 33

All are direct branches of aortic arch except

- A. brachiocephalic trunk
- B. left common carotid**
- C. right subclavian
- D. none

Correct Answer: C. right subclavian

Explanation: The arch of the aorta directly gives rise to three major branches: the **brachiocephalic trunk**, the **left common carotid**, and the **left subclavian** artery. The **right subclavian** is *not* a direct branch of the aortic arch; rather, it arises as a branch of the brachiocephalic trunk on the right side.

[24/09/2026 9:44 AM] Imaanaa Harmee: Question 34

A 23-year-old male injured in an industrial explosion was found to have multiple small metal fragments in his thoracic cavity. Since the pericardium was torn inferiorly, the surgeon began to explore for fragments in the pericardial sac. Slipping her hand under the heart apex, she slid her fingers upward and to the right within the sac until they were stopped by the cul-de-sac formed by the pericardial reflection near the base of the heart. Her fingertips were then in the:

- A. Coronary sulcus
- B. Costomediastinal recess**
- C. Oblique sinus
- D. Transverse sinus

Correct Answer: C. Oblique sinus

Explanation: The **oblique sinus** is a blind-ended recess (cul-de-sac) of the pericardial cavity located posteriorly behind the heart. It is formed by the reflection of the serous pericardium around the pulmonary veins and the inferior vena cava. If a surgeon slips her hand under the

apex and slides her fingers upward and to the right behind the heart, they will enter and be stopped by the walls of this oblique sinus.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 35

Which pair is incorrect

- A. Condyle: Rounded articular projection
- B. Trochanter: Very large, blunt projection
- C. Tuberosity: large rounded or roughened projection
- D. Fissure: Cleft
- E. None

Correct Answer: E. None

E

Explanation: All of the pairing terms are anatomically correct definitions of bone markings:

✓ **Condyle:** A rounded articular projection.

✓ **Trochanter:** A very large, blunt, irregularly shaped projection (uniquely found on the femur).

✓ **Tuberosity:** A large, rounded, or roughened projection.

✓ **Fissure:** A cleft or narrow, slit-like opening.

Since all of these definitions are correct, **None** of them is incorrect.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 36

_____ is butterfly-shaped bone with outstretched wings

- A. Sphenoid
- B. Parietal
- C. Frontal
- D. Hyoid

Correct Answer: A. Sphenoid

Explanation: The **sphenoid bone** is classically described as a butterfly-shaped bone situated at the base of the skull. This unique shape is formed by its central body and **outstretched lateral projections** known as the **greater and lesser wings**.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 37

Which of the following muscle is named based on its location

- A. Temporalis
- B. Deltoid
- C. Intercostal muscles
- D. A and C

Correct Answer: D. A and C (Temporalis and Intercostal muscles)

Explanation:

◇ **Temporalis** is named directly after its **location** in the temporal fossa overlying the temporal bone of the skull.

◇ **Intercostal muscles** are named for their **anatomical position** between the ribs (*inter* = between, *costal* = ribs).

◇ *Deltoid* (Option B) is named based on its triangular **shape** (resembling the Greek letter Delta) rather than its location.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 38

Which of the following muscle is the longest in the human body

- A. Gluteal Maximus
- B. Sartorius
- C. Deltoid
- D. Quadriceps femoris

Correct Answer: B. Sartorius

Explanation: The **sartorius muscle** is a long, thin, band-like muscle that spans two joints. It originates at the anterior-superior iliac spine of the hip and inserts onto the upper medial side of the tibia. It is classically recognized as the **longest muscle in the human body**.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 39

All skeletal muscles have all except

- A. Origin
- B. Action
- C. Nerve
- D. None

Correct Answer: D. None

Explanation: By anatomical definition, **all skeletal muscles** possess an **origin** (the more fixed, proximal attachment), a **nerve supply** that controls voluntary contractions, and a specific **action** (or movement) that is produced when the muscle contracts. Because all skeletal muscles have all three of these features, there is no exception (**None** of the options is false).

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 40

Muscles of mastication are innervated by

- A. Facial nerve
- B. Trigeminal
- C. Musculocutaneous
- D. Radial

Correct Answer: B. Trigeminal

Explanation: All four muscles of mastication (the temporalis, masseter, lateral pterygoid, and medial pterygoid) are innervated by the **mandibular division of the trigeminal nerve (CN V3)**. These muscles develop from the first pharyngeal (branchial) arch, which is supplied exclusively by the trigeminal nerve.

Exam 2017

Question 1

Which is false

- A. Nervous system is the smallest in size from all systems in the human body
- B. PNS has less regeneration capacity compared to CNS
- C. Unipolar neurons are found in the human body
- D. Brain size is directly proportional to intelligence
- E. All except A

Correct Answer: E. All except A

◇ **Why B is false:** The **Peripheral Nervous System (PNS)** has a significantly **higher regeneration capacity** compared to the **Central Nervous System (CNS)**. Axons in the PNS can regenerate under favorable conditions with the help of Schwann cells, whereas the CNS has extremely limited to no regenerative capacity due to oligodendrocytes and glial scarring.

◇ **Why C is false:** True unipolar neurons are not typically found in the human adult body. Instead, they are **pseudounipolar** (sensory neurons of the PNS found in the spinal and cranial ganglia), which begin as bipolar during embryological development before fusing into a single process.

◇ **Why D is false:** There is no direct, scientific correlation showing that brain size is directly proportional to human intelligence.
Since **B, C, and D are all false**, **E** is the correct choice.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 2

All are not true except

- A. About half of the total volume of human brain is composed of Neurons cells.
- B. Neuroglia are excitable cells
- C. Neuroglia are larger than neurons
- D. Neuroglia can regenerate them selves
- E. None

Correct Answer: D. Neuroglia can regenerate themselves

◇ **A is incorrect:** About half of the total volume of the brain is composed of **neuroglia (glial cells)**, not neurons.

◇ **B is incorrect:** Neurons are the excitable cells of the nervous system; **neuroglia are non-excitabile** and do not propagate action potentials.

◇ **C is incorrect:** Neuroglia are generally **much smaller** than neurons, although they outnumber them significantly.

◇ **D is correct:** Unlike neurons, neuroglia retain their **mitotic ability** and can readily divide and regenerate themselves throughout life (which is also why most brain tumors, like gliomas, originate from glial cells).

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 3

_____ is the main phagocytic cells in the CNS.

Answer: Microglia (or microglial cells)

Explanation: **Microglia** are specialized macrophage-like cells that serve as the primary active immune defense and **main phagocytic cells** within the Central Nervous System, clearing out waste products, damaged neurons, and pathogens.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 4

_____ the ventricle lines the midbrain

Correct Answer: Cerebral aqueduct (aqueduct of Sylvius)

Explanation: The **cerebral aqueduct** is the narrow, fluid-filled canal of the ventricular system that runs directly through the **midbrain**, connecting the third ventricle of the diencephalon to the fourth ventricle of the hindbrain.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 5

Which is false

- A. Duramatter is double layered in cerebrum
- B. Dural layers are fused together at all places
- C. Duramatter extends to 2nd sacral vertebrae
- D. Arachnoid mater is avascular
- E. the denticulate ligaments attaches the spinal cord to the dura mater

Correct Answer: B. Dural layers are fused together at all places

◇ **Explanation:** cranial dura mater consists of two layers (an outer periosteal layer and an inner meningeal layer). These two layers are **not** fused together at all places; they separate at specific locations to form the **dural venous sinuses** (such as the superior sagittal sinus).

The other statements are anatomically true:

- ✓ The spinal dural sac and subdural/subarachnoid spaces extend inferiorly to the level of the **second sacral vertebra (S2)**.
- ✓ The arachnoid mater is a delicate, **avascular** membrane.
- ✓ The **denticulate ligaments** are lateral extensions of the pia mater that traverse the arachnoid space to anchor the spinal cord to the dura mater.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 6

All are false except

- A. falx cerebri separates cerebellum
- B. graymatter is medullary in position in cerebrum in contrast to spinal cord
- C. The lateral sulcus separates the frontal and parietal lobe from the occipital lobe
- D. Thalamus helps for relay center for all sensory impulses
- E. none

Correct Answer: E. none

◇ **Why A is false:** The **falx cerebri** is a double layer of dura mater that lies between and separates the **cerebral hemispheres**. It is the *falx cerebelli* that projects between the cerebellar hemispheres, and the *tentorium cerebelli* that covers and supports the occipital lobes over the cerebellum.

◇ **Why B is false:** In the cerebral hemispheres (cerebrum), gray matter is **cortical** (located on the outer surface), whereas in the spinal cord, the gray matter is **centrally located**. Therefore, gray matter is cortical in the cerebrum, not medullary.

◇ **Why C is false:** The lateral sulcus (or lateral fissure) is the landmark that separates the frontal and parietal lobes from the underlying **temporal lobe**, not the posterior occipital lobe.

◇ **Why D is false:** While the thalamus is the major sensory gateway of the brain, it acts as a relay center for all sensory impulses **except olfactory (smell) sensations**, which bypass the thalamus to project directly to the primary olfactory cortex.

◇ Since all the choices above are false, **none** is the correct selection.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 7

_____ a broad region in the left temporal and parietal lobes which interprets the meaning of speech by recognizing spoken words.

Correct Answer: Wernicke's area

Explanation: Located in the posterior portion of the left temporal and parietal lobes (the dominant hemisphere), **Wernicke's area** is the specialized sensory speech region responsible for interpreting the meaning of language, translating spoken words into coherent thoughts, and recognizing speech patterns.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 8

Which is true

- A. Lateral horn of spinal cord is found in cervical region

- B. Olfactory nerve arise from brain stem
- C. Trigeminal nerve is the longest cranial nerve
- D. Glossopharyngeal nerve contain autonomic part
- E. Sciatic nerve is the largest cranial nerve in the human body which arise from sacral plexus

Correct Answer: D. Glossopharyngeal nerve contain autonomic part



Why D is true: The **glossopharyngeal nerve (CN IX)** contains a **general visceral efferent (GVE)** component. This autonomic portion carries preganglionic parasympathetic fibers that synapse in the otic ganglion to provide secretomotor supply to the parotid salivary gland.

Why the others are false:

◇ **A is false:** The lateral horn of the spinal cord (which houses preganglionic sympathetic cell bodies) is only present in the **thoracolumbar segments (T1 to L2)**, not the cervical region.

◇ **B is false:** The olfactory nerve (CN I) consists of special sensory bundles that pass through the cribriform plate of the ethmoid bone to terminate directly in the olfactory bulb of the foregut/cerebrum, meaning it does not arise from the brainstem.

◇ **C is false:** The **vagus nerve (CN X)** is the longest cranial nerve, extending from the brainstem all the way down into the thoracic and abdominal cavities.

◇ **E is false:** The sciatic nerve arises from the sacral plexus and is the largest nerve in the body, but it is a **spinal/peripheral nerve**, not a cranial nerve.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 9

Which is true

- A. Laryngopharynx begins at the level of the hyoid bone
- B. It opens in to esophagus ventrally
- C. cricoid cartilage is single of laryngeal cartilage
- D. All
- E. All except B

Correct Answer: E. All except B

◇ **Why A is true:** The laryngopharynx extends from the upper border of the epiglottis (which lies at the level of the hyoid bone) down to the lower border of the cricoid cartilage.

◇ **Why B is false:** The pharynx is located posterior (dorsal) to the larynx, and the esophagus runs directly behind (posterior to) the trachea. Therefore, the laryngopharynx opens into the

esophagus **dorsally** (posteriorly), not ventrally (anteriorly).

♦ **Why C is true:** The **cricoid cartilage** is a single (unpaired) signet-ring-shaped hyaline cartilage of the larynx.

Because A and C are true but B is false, **E** is the correct answer.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 10

Why thyroid cartilage is larger in males compared to females?

The thyroid cartilage is significantly larger in males than in females due to the influence of **androgens (testosterone)** during puberty.

During male adolescence, increased testosterone secretion stimulates rapid growth and expansion of the laryngeal skeleton. This causes the two laminae of the hyaline thyroid cartilage to fuse at a much sharper angle in males (approximately 90°) than in females (approximately 120°). This acute angulation pushes the cartilage forward, creating a prominent midline thyroid projection known as the **laryngeal prominence** or "**Adam's apple**". This anatomical expansion also increases the length of the vocal cords inside the larynx, resulting in the characteristic deepening of the male voice.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 11

Tracheal mucosa is lined by _____ epithelium.

Correct Answer: pseudostratified ciliated columnar (or simply **pseudostratified ciliated**) epithelium.

Explanation: The mucosa of the trachea (along with the larger bronchi) is classically lined by **pseudostratified ciliated columnar epithelium with goblet cells**. This is also widely referred to as **respiratory epithelium**, which functions to filter, humidify, and warm the air as it passes through.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 12

Which is true

- A. Right lung is broader than left
- B. Left lung has cardiac notch
- C. Left lung has transverse fissure
- D. Hilum is found in both lungs
- E. All except C

Correct Answer: E. All except C

Explanation:

◇ **A is true:** The **right lung** is shorter but **wider (broader) and heavier** than the left lung due to the presence of the liver's large right dome.

◇ **B is true:** The anterior border of the superior lobe of the **left lung** features a deep indentation known as the **cardiac notch**.

◇ **C is false:** Only the right lung is divided by both an oblique and a horizontal (transverse/accessory) fissure. The **left lung has only an oblique fissure**, meaning it lacks a transverse fissure.

◇ **D is true:** The **hilum (hilus)** is the entry/exit gateway for bronchi and pulmonary vessels, and it is a structural feature **found in both lungs**.

Since A, B, and D are true, the correct choice is **E**.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 13

As the branching becomes extensive in the bronchial tree, smooth muscle is increasing

A. True

B. false

Correct Answer: A. True

Explanation: As the bronchial tree branches deeper into the lungs, the supportive cartilage rings progressively decrease and disappear (disappearing completely at the bronchiole level). Concurrently, the relative proportion of **smooth muscle** in the walls of the airways **increases** relative to their total diameter to regulate airway resistance.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 14

All are true except

A. Mucosal layer of GIT help for secretion of hormones

B. lamina propria is the innermost layer of mucosa which is loose connective tissue

C. Meissner's plexus are located in muscularis externa

D. Dodinum has both serosa and adventia

E. All except A

Correct Answer: C. Meissner's plexus are located in muscularis externa

Explanation: This statement is **false** (making it the correct "except" choice) because the **submucosal (Meissner's) plexus** is located within the **submucosa** layer of the gastrointestinal tract. It is the **myenteric (Auerbach's) plexus** that is located within the **muscularis externa** (specifically between the circular and longitudinal muscle layers).

◇ *Other options are true:* Enteroendocrine cells in the mucosa secrete hormones like gastrin; the duodenum has both serosa (on its peritonealized first part/anterior surface) and adventitia (on its retroperitoneal surfaces).

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 15
_____ is the largest salivary gland which secretes saliva at _____.

Correct Answers: Parotid gland; opposite the second upper molar tooth

Explanation:

The **parotid gland** is the largest of the three salivary glands. It has a specialized excretory duct (the parotid/Stensen's duct) that crosses the masseter muscle, pierces the buccinator muscle, and opens into the vestibule of the oral cavity **opposite the second upper molar tooth**.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 16

The dentin of the root is covered by _____.

Correct Answer: cement (or cementum)

Explanation: The anatomy of a tooth consists of a crown, a neck, and a root. While the crown is covered by enamel, the dentin of the root is covered with **cement** (cementum), which is connected to the alveolar bone by the periodontal ligament.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 17

Which is true

- A. Stomach has triple layered muscularis mucosae layer
- B. Liver is the largest organ in the human body which is located in the right hypochondriac region
- C. Colon has smooth lumen compared to ilium
- D. Large intestine is more fixed than small due to its length
- E. Ilium is the longest part of small intestine which has Peyer's patches at its proximal end

Correct Answer: C. Colon has smooth lumen compared to ileum

Explanation:

◇ **C is true** because the mucosal lining of the colon lacks plicae circulares (circular folds) and villi, giving its internal lumen a much **smoother** appearance compared to the highly folded, villi-dense ileum.

Why the other options are false:

◇ **A is false:** The stomach has three smooth muscle layers in its **muscularis externa** (inner oblique, middle circular, outer longitudinal), not its *muscularis mucosae*.

◇ **B is false:** The **skin** is the largest organ of the human body. The liver is the largest *visceral (internal) organ* and the largest gland.

◇ **D is false:** The large intestine is more fixed because key parts (ascending and descending colon) are retroperitoneal, not "due to its length."

◇ **E is false:** The intestinal section is spelled **ileum** ("ilium" is the pelvic bone). Furthermore, Peyer's patches are located at the **distal (lower) end** of the ileum, not the proximal end.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 18

Which of the following is NOT contained in the superior mediastinum?

- A. Great vessels and their branches
- B. Azygos and hemiazygos veins
- C. Trachea
- D. Esophagus

Correct Answer: B. Azygos and hemiazygos veins

Explanation:

◇ The **superior mediastinum** contains the great vessels (and their branches), the trachea, and the esophagus.

◇ The **azygos and hemiazygos veins** run in the **posterior mediastinum** (with the arch of the azygos vein looping into the middle mediastinum to enter the superior vena cava).

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 19

The attending faculty in the coronary intensive care unit (ICU) demonstrates to his students a normal heart examination. The first heart sound is produced by near-simultaneous closure of which of the following valves?

- A. Aortic and tricuspid
- B. Aortic and pulmonary

- C. Tricuspid and mitral
- D. Mitral and pulmonary

Correct Answer: C. Tricuspid and mitral

Explanation: The **first heart sound ("lub")** is produced by the near-simultaneous closure of the atrioventricular (AV) valves—the **tricuspid and mitral (bicuspid) valves**—at the onset of ventricular contraction (systole).

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 20

A 27-year-old cardiac patient with an irregular heartbeat visits her doctor's office for examination. Where should the physician place the stethoscope to listen to the sound of the Aortic valve?

- A. Over the medial end of the second left intercostal space
- B. Over the medial end of the second right intercostal space
- C. In the left fourth intercostal space at the midclavicular line
- D. In the left fifth intercostal space at the midclavicular line

Correct Answer: B. Over the medial end of the second right intercostal space

Explanation: Although the anatomical location of the aortic valve lies behind the left half of the sternum, the sound of the valve is carried by the outflow of blood and is **most audible over the right second intercostal space** just lateral to the sternum (which is the medial end of the space).

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 21

Select the arteries not involved in forming circle of Willis?

- A. Posterior communicating arteries
- B. Posterior cerebral arteries
- C. Anterior cerebral arteries
- D. Anterior communicating arteries
- E. Middle cerebral arteries

Correct Answer: E. Middle cerebral arteries

Explanation:

◇ The **circle of Willis** (circulus arteriosus) is an important polygonal anastomosis at the base of the brain.

◇ it is formed by the **posterior cerebral, posterior communicating, internal carotid, anterior cerebral, and anterior communicating arteries**.

◇ The **middle cerebral artery** (the larger terminal branch of the internal carotid) arises from this area but is **not** structurally involved in forming the closed anastomotic loop of the circle itself.

[24/09/2026 9:49 AM] Imaanaa Harmee: Question 22

The most (80%) of the brain supplies by?

- A. Carotid arteries
- B. Vertebral artery
- C. Basal arteries
- D. ophthalmic artery

Correct Answer: A. Carotid arteries

Explanation:

◇ The brain receives its blood supply from two main arterial systems: the **internal carotid arteries** (anterior circulation) and the **vertebral arteries** (posterior circulation).

◇ The **carotid system** supplies approximately **80% of the brain's blood flow** (encompassing the cerebral hemispheres and forebrain), while the vertebrobasilar system supplies the remaining 20% (including the brainstem, cerebellum, and occipital lobes).

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 23

The posterior tibial artery enters the sole and bifurcates into:

- A. lateral and medial plantar arteries
- B. posterior tibial and dorsalis pedis
- C. Anterior Tibial and Posterior Tibial
- D. All

Correct Answer: A. lateral and medial plantar arteries

Explanation:

✓ the **posterior tibial artery** descends through the posterior compartment of the leg, passes behind the medial malleolus, and "**terminates beneath the flexor retinaculum by dividing into the medial and lateral plantar arteries**".

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 24

Which of the following is conducting arteries?

- A. Femoral arteries
- B. Radial arteries
- C. Brachiocephalic
- D. Popliteal arteries
- E. A and C

Correct Answer: C. Brachiocephalic

Explanation:

◇ Blood vessels are divided into elastic (conducting) arteries, muscular (distributing) arteries, and arterioles.

◇ **Conducting (elastic) arteries** are large-caliber vessels close to the heart with walls rich in elastic fibers that help maintain continuous blood pressure during diastole. Examples include the aorta, **brachiocephalic trunk**, subclavian, and common carotid arteries.

◇ *Femoral, radial, and popliteal arteries* are classified as muscular (distributing) arteries.

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 25

Select the correct one?

- A. Diaphragmatic (inferior) surface of the heart formed by left ventricle (left 2/3) and right atrium (right 1/3)
- B. Sternocostal (anterior) surface of the heart formed by right atrium and right ventricle
- C. Diaphragmatic (inferior) surface of the heart formed by left ventricle (left 1/3) and right ventricle (right 2/3)
- D. None of the above

Correct Answer: D. None of the above

Explanation:

◇ **Why A is incorrect:** The diaphragmatic (inferior) surface of the heart is formed by the **left ventricle (left 2/3)** and **right ventricle (right 1/3)**, *not* the right atrium.

◇ **Why B is incorrect:** The sternocostal (anterior) surface of the heart is formed **primarily by the right ventricle**, but also includes portions of the right atrium and the left ventricle. Describing it as formed only by the right atrium and right ventricle is incomplete and anatomically incorrect.

◇ **Why C is incorrect:** This option reverses the correct ventricular proportions of the diaphragmatic surface (it should be left ventricle 2/3 and right ventricle 1/3).

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 26

The common site of vein puncture?

- A. Cubital vein
- B. Basilic vein
- C. Cephalic vein
- D. Femoral vein

Correct Answer: A. Cubital vein

Explanation: Venipuncture in the upper limb is classically performed on the **median cubital vein** (referred to here as the cubital vein). This vein lies superficially in the cubital fossa, separating from the deeper brachial artery by the protective bicipital aponeurosis. While the cephalic and basilic veins can also be used, the median cubital vein is the most common site.

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 27

The apex of the heart is formed by:

- A. Right ventricle
- B. left atrium
- C. left ventricle
- D. Right atrium

Correct Answer: C. left ventricle

Explanation: The apex of the heart is the blunt, rounded, inferolateral extremity of the organ. It is formed entirely by the **left ventricle** and is clinically located in the left fifth intercostal space, slightly medial to the midclavicular line.

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 28

Which of the following is the correct pass of urine coming out of kidneys?

- A. PCT- DCT - Loop of Henle - Collecting duct - Minor calyx- Major calyx -Renal Pelvis
- B. PCT- Loop of Henle- DCT- Collecting duct- Major calyx- Minor calyx -Renal Pelvis
- C. PCT- Loop of Henle- DCT- Collecting duct - Minor calyx- Major calyx -Renal Pelvis
- D. PCT- Loop of Henle- DCT -Collecting duct - Major calyx--Renal Pelvis- Minor calyx
- E. None

Correct Answer: C. PCT- Loop of Henle- DCT- Collecting duct - Minor calyx- Major calyx -Renal Pelvis

Explanation:

- ◇ The flow of filtrate through the functional units of the kidney (the nephrons) begins in the

renal corpuscle, moving sequentially through the **proximal convoluted tubule (PCT)**, **Henle's loop**, and the **distal convoluted tubule (DCT)**.

◇ From the DCT, the fluid enters the **collecting duct (collecting tubules)**, which open at the renal papillae.

◇ From the papillae, urine drains into the **minor calyces**, which merge to form **major calyces**, and finally empties into the dilated **renal pelvis** before traveling down the ureter.

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 29

The tuft of capillaries in the renal corpuscle is called the

- A. Podocytes.
- B. Glomerulus.
- C. Calyx.
- D. Renal pyramid.
- E. Renal sinus.

Correct Answer: B. Glomerulus.

Explanation: The renal corpuscle (the filtration unit of the nephron) consists of a **glomerulus**—which is a specialized **tuft of capillaries**—surrounded by a double-walled glomerular (Bowman's) capsule.

[24/09/2026 9:54 AM] Imaanaa Harmee: Question 30

The _____ collect(s) urine, which drains continuously from the papillae; the urine is then emptied into the _____.

- A. renal pyramids; renal pelvis
- B. calyces; renal pelvis
- C. renal pelvis; calyces
- D. renal pyramids; calyces

Correct Answer

: B. calyces; renal pelvis

Explanation: The cup-shaped **minor and major calyces** directly receive and collect urine as it continuously drains from the openings on the renal papillae (pyramids). This collected urine is then funneled and emptied into the **renal pelvis**, which represents the upper expanded portion of the ureter.

